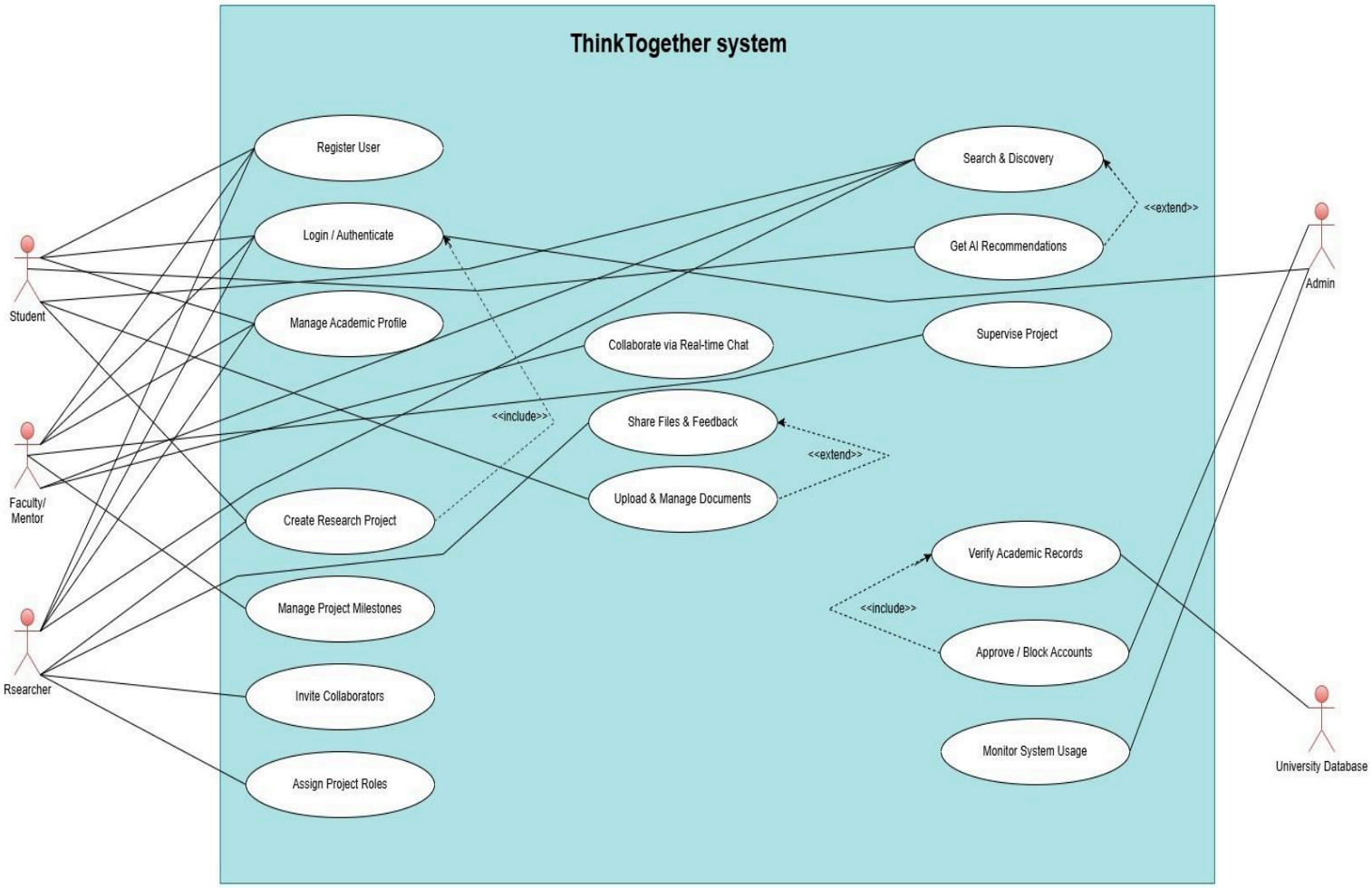
Use case Diagram



Adnan Alam	ID: 20245103051
SM. Azman Sikder Durjay	ID: 20245103133
Md. Mostar-Shid Billah	ID : 20245103139
Dipta Dey	ID: 20245103143
Shudipto Ghosh	ID: 20245103151

Use Case Specification

UC-01: Register User

Title	Register User
Description	Allows a new user to create an account using university credentials.
Actor and Interfaces	Student, Faculty, Researcher, University Database.
Initial status and Precondition	The user is not registered in the system; must have a valid university email.
Basic Flow **Step 1:** User enters registration details (Email, ID). **Step 2:** System sends a verification request to the University Database. **Step 3:** System confirms validity and prompts user to set a password. **Step 4:** User completes profile information.	
Post condition : Account is created and status is set to "Pending Admin Approval".	
Alternative Flow	**A1:** University Database returns "Record Not Found"; system displays error.

UC-02: Login / Authenticate

Title	Login / Authenticate
Description	Grants a registered user access to the platform.
Actor and Interfaces	Student, Faculty, Researcher, Admin.
Initial status and Precondition	Users must have an active, verified account.
Basic Flow **Step 1:** User enters university email/ID and password. **Step 2:** System validates credentials against the User Store. **Step 3:** System redirects user to the personalized dashboard.	
Post condition : User is granted session access.	
Alternative Flow	**A1:** Incorrect password; system prompts for account recovery.

UC-03: Manage Academic Profile

Title	Manage Academic Profile
Description	Allows users to update educational details, CGPA, and research interests.
Actor and Interfaces	Student, Faculty, Researcher.
Initial status and Precondition	Users must be logged in.
Basic Flow **Step 1:** User navigates to profile settings. **Step 2:** User updates academic information (publications, research interests, etc.). **Step 3:** User saves changes.	
Post condition : Profile is updated in the User Store.	
Alternative Flow	**A1:** User inputs invalid CGPA format; system requests correction.

UC-04: Create Research Project

Title	Create Research Project
Description	Initializes a new research workspace for team collaboration.
Actor and Interfaces	Student, Faculty, Researcher.
Initial status and Precondition	Users must be logged in.
Basic Flow **Step 1:** User clicks "Create Project". **Step 2:** User inputs title, domain, and objectives. **Step 3:** System generates a unique Project ID and workspace.	
Post condition : New project is stored in the Project Store.	
Alternative Flow	N/A.

UC-05: Manage Project Milestones

Title	Manage Project Milestones
Description	Sets deadlines and tracks the visual progress of a project.
Actor and Interfaces	Project Lead, Faculty/Mentor.
Initial status and Precondition	An active project must exist.
Basic Flow **Step 1:** Lead/Mentor accesses the project dashboard. **Step 2:** User defines milestones and assigns deadlines. **Step 3:** System updates the progress tracker.	
Post condition : Project timeline is updated and visible to all members.	
Alternative Flow	N/A.

UC-06: Invite Collaborators

Title	Invite Collaborators
Description	Sends requests to other users to join a specific research project.
Actor and Interfaces	Student, Researcher.
Initial status and Precondition	User must be the Project Lead
Basic Flow **Step 1**: Lead searches for a user by name or interest. **Step 2**: Lead clicks "Invite" on the user profile. **Step 3**: System sends a notification to the invited user.	
Post condition : Invitation status is tracked in the system.	
Alternative Flow	N/A.

UC-07: Assign Project Roles

Title	Assign Project Roles
Description	Defines specific permissions (Lead, Member, Mentor) within a project.
Actor and Interfaces	Project Lead, Faculty/Mentor.
Initial status and Precondition	Multiple users must be part of the project.
Basic Flow **Step 1:** Lead opens the "Members" tab. **Step 2:** Lead selects a member and chooses a role from the dropdown menu. **Step 3:** System updates permissions for that member.	
Post condition : Role-based access is applied to project resources.	
Alternative Flow	N/A.

UC-08: Collaborate via Real-time Chat

Title	Collaborate via Real-time Chat
Description	Provides instant messaging for group-based communication.
Actor and Interfaces	Student, Faculty, Researcher.
Initial status and Precondition	Users must be an active member of the project group.
Basic Flow **Step 1**: The user opens the project chat interface. **Step 2**: User enters a message. **Step 3**: System broadcasts the message to all group members.	
Post condition : Chat history is stored in the Log Store.	
Alternative Flow	N/A.

UC-09: Share Files & Feedback

Title	Share Files & Feedback
Description	Allows members to comment on documents and exchange feedback
Actor and Interfaces	Project Members, Faculty/Mentor.
Initial status and Precondition	A document must be uploaded to the project workspace.
Basic Flow **Step 1:** User selects an uploaded document. **Step 2:** User adds a comment or structured feedback. **Step 3:** System notifies the author of the feedback.	
Post condition : Feedback is appended to the document metadata	
Alternative Flow	N/A.

UC-10: Upload & Manage Documents

Title	Upload & Manage Documents
Description	Stores research papers and files with version history
Actor and Interfaces	Student, Researcher, Faculty.
Initial status and Precondition	User must have upload permissions for the project
Basic Flow **Step 1:** User selects "Upload".. **Step 2:** User chooses file and sets visibility (Public/Private). **Step 3:** System stores the file in the Document Repository and logs the version	
Post condition : File is accessible to the designated project team.	
Alternative Flow	**A1:** File format is not supported; system rejects upload.

UC-11: Search & Discovery

Title	Search & Discovery
Description	Allows users to find peers, projects, or papers based on research interests.
Actor and Interfaces	All Users.
Initial status and Precondition	The user is logged in.
Basic Flow **Step 1:** The user enters keywords in the search engine. **Step 2:** User filters results by domain or university. **Step 3:** System displays matching profiles or research work.	
Post condition : N/A.	
Alternative Flow	N/A.

UC-12: Get AI Recommendations

Title	Get AI Recommendations
Description	The system automatically suggests mentors or research topics (Extends UC-11).
Actor and Interfaces	Discovery Engine, All Users.
Initial status and Precondition	The user has a complete academic profile with research interests.
Basic Flow **Step 1:** AI engine analyzes user's recent searches and profile interests. **Step 2:** System generates a "Recommended for You" list. **Step 3:** User views suggested collaborators or topics.	
Post condition : N/A.	
Alternative Flow	N/A.

UC-13: Supervise Project

Title	Supervise Project
Description	Faculty review of student research and project progress.
Actor and Interfaces	Faculty/Mentor
Initial status and Precondition	Faculty must be assigned as a Mentor to the project
Basic Flow **Step 1:** Mentor views the project dashboard **Step 2:** Mentor reviews documents and milestones **Step 3:** Mentor provides approval or formal feedback.	
Post condition : Supervision log is updated.	
Alternative Flow	N/A.

UC-14: Verify Academic Records

Title	Verify Academic Records
Description	Validates the authenticity of student credentials with the university.
Actor and Interfaces	Admin, University Database.
Initial status and Precondition	System or Admin triggers a verification check during registration.
Basic Flow **Step 1:** System sends user data to the University Database API. **Step 2:** Database verifies ID and records (CGPA, enrollment). **Step 3:** Database sends a confirmation/rejection status back to the system.	
Post condition : User status is verified.	
Alternative Flow	**A1:** Connection to Database fails; Admin is notified to verify manually.

UC-15: Approve / Block Accounts

Title	Approve / Block Accounts
Description	Administrative control over system access based on verification.
Actor and Interfaces	Admin.
Initial status and Precondition	The verification process (UC-14) must be complete.
Basic Flow **Step 1:** Admin reviews the "Pending Users" list. **Step 2:** Admin selects "Approve" or "Block" for a specific user. **Step 3:** System updates the account status and notifies the user.	
Post condition : User access is granted or revoked.	
Alternative Flow	**N/A.**

UC-16: Monitor System Usage

Title	Monitor System Usage
Description	Generates reports on platform activity and usage statistics.
Actor and Interfaces	Admin
Initial status and Precondition	Admin is logged into the management console
Basic Flow **Step 1:** Admin requests a system report (e.g., active projects). **Step 2:** System fetches data from the Log Store and Analytics Engine. **Step 3:** System generates a visual report for the Admin.	
Post condition : N/A	
Alternative Flow	N/A

Adnan Alam	20245103051
S.M Azman Sikder Durjay	20245103133
Mostar-Shid Billah	20245103139
Dipta Dey	20245103143
Shudipto Ghosh	20245103151